



Skill: knowledge

Atomic number	Symbol	Name	Atomic weight	Atomic radius*
1	H	hydrogen	1	37
2	He	helium	4	90
3	Li	lithium	7	125
4	Be	beryllium	9	90
5	B	boron	11	82
6	C	carbon	12	77
7	N	nitrogen	14	75
8	O	oxygen	16	73
9	F	fluorine	19	72
10	Ne	neon	20	71
11	Na	sodium	23	154
12	Mg	magnesium	24	136
13	Al	aluminum	27	118
14	Si	silicon	28	110
15	P	phosphorus	31	106
16	S	sulfur	32	102
17	Cl	chlorine	35	99
18	Ar	argon	40	95
19	K	potassium	39	203
20	Ca	calcium	40	174
21	Sc	scandium	45	144
22	Ti	titanium	48	132
23	V	vanadium	51	122
24	Cr	chromium	52	118
25	Mn	manganese	55	117
26	Fe	iron	56	117
27	Co	cobalt	59	116
28	Ni	nickel	59	115
29	Cu	copper	64	117
30	Zn	zinc	65	125
31	Ga	Gallium	70	126
32	Ge	germanium	73	122
33	As	arsenic	75	120
34	Se	selenium	79	117
35	Br	bromine	80	114
36	Kr	krypton	84	112
37	Rb	rubidium	85	216
38	Sr	strontium	88	191
39	Y	yttrium	89	162
40	Zr	zirconium	91	145
41	Nb	niobium	93	134
42	Mo	molybdenum	96	130
43	Tc	technetium	98	127
44	Ru	ruthenium	101	134
45	Rh	rhodium	103	125
46	Pd	palladium	106	128
47	Ag	silver	108	134
48	Cd	cadmium	112	148
49	In	indium	115	144
50	Sn	tin	119	141
51	Sb	antimony	122	140
52	Te	tellurium	128	136
53	I	iodine	127	133
54	Xe	xenon	131	131
55	Cs	cesium	133	235
56	Ba	barium	137	198
57	La	lanthanum	139	169
58	Ce	cerium	140	165
59	Pr	praseodymium	141	165

Atomic number	Symbol	Name	Atomic weight	Atomic radius*
60	Nd	neodymium	144	164
61	Pm	promethium	145	163
62	Sm	samarium	150	162
63	Eu	europium	152	185
64	Gd	gadolinium	157	161
65	Tb	terbium	159	159
66	Dy	dysprosium	163	159
67	Ho	holmium	165	158
68	Er	erbium	167	157
69	Tm	thulium	169	156
70	Yb	ytterbium	173	144
71	Lu	lutetium	175	134
72	Hf	hafnium	178	136
73	Ta	tantalum	181	128
74	W	tungsten	184	126
75	Re	rhenium	186	127
76	Os	osmium	190	128
77	Ir	iridium	192	126
78	Pt	platinum	195	129
79	Au	gold	197	134
80	Hg	mercury	201	149
81	Tl	thallium	204	148
82	Pb	lead	207	146
83	Bi	bismuth	209	145
84	Po	polonium	209	
85	At	astatine	210	
86	Rn	radon	222	
87	Fr	francium	223	
88	Ra	radium	226	
89	Ac	actinium	227	
90	Th	thorium	232	
91	Pa	protactinium	231	
92	U	uranium	238	
93	Np	neptunium	237	
94	Pu	plutonium	244	
95	Am	americium	243	
96	Cm	curium	247	
97	Bk	berkelium	247	
98	Cf	californium	251	
99	Es	einsteinium	252	
100	Fm	fermium	257	
101	Md	mendelevium	258	
102	No	nobelium	259	
103	Lr	lawrencium	262	
104	Rf	rutherfordium	261	
105	Db	dubnium	262	
106	Sg	seaborgium	266	
107	Bh	bohrium	264	
108	Hs	hassium	277	
109	Mt	meitnerium	268	
110	Ds	darmstadtium	271	
111	Rg	roentgenium	272	
112	Uub	ununbium	285	

*Atomic radii are in picometres (pm).

The elements — page 2

Name: _____

Questions

Use the data on the previous page to answer these questions.

1 Give the symbols for the following elements:

- tungsten _____
- sodium _____
- zinc _____
- einsteinium _____
- tin _____
- boron _____
- cerium _____
- radon _____

2 Give the names for the following elements:

- Am _____
- Uub _____
- Ne _____
- Ar _____
- Cd _____
- O _____
- H _____
- Si _____
- S _____

3 Identify the following atoms from the table, giving their names and symbols:

- a the atom with an atomic number of 88

- b the atom with an atomic weight of 39

- c the atom with an atomic radius of 133

- d the atom with the symbol Hg

- e the atom with an atomic number of 16

- f the atom with the symbol He

- g the atom with atomic weight 108

4 The atomic weight of an atom is its mass compared with that of a carbon atom. If an atom has an atomic weight less than carbon it is lighter than carbon. If an atom has an atomic weight larger than carbon it is heavier than carbon.

a State the atomic weight of a carbon atom.

b Identify two atoms that are lighter than carbon.

c Identify two atoms that are heavier than carbon.

d Describe what happens to the atomic weight as the atomic number increases.

e Calculate how many times heavier a magnesium atom is than carbon.

f Calculate how many times lighter a helium atom is than carbon.

5 The atomic radius gives an indication of the size of an atom. The table does not give the atomic radius for all atoms. Use the data shown to give the name and symbol for the:

a smallest atom _____

b largest atom _____

The periodic table—page 2

Name: _____

Questions

- 1 On the periodic table, colour each of the following in different colours:
 - a metals
 - b non-metals
 - c semi-metals (metalloids).
- 2 Identify which group consists of all gases.

- 3 What other name is given to the group VII elements?

- 4 State the atomic number for these elements:
 - a platinum _____
 - b gold _____
 - c oxygen _____
 - d magnesium _____
 - e californium _____
 - f bromine _____
 - g krypton _____
- 5 State the name for atoms with the following atomic numbers:
 - a 6 _____
 - b 3 _____
 - c 40 _____
 - d 13 _____
 - e 92 _____
 - f 56 _____
 - g 1 _____
- 6 a Identify three elements that start with a C.

 - b State the symbols for the elements you have chosen.

 - c Explain why some elements have one letter for their symbol and others have two.

- 7 Outline a reason why elements are placed in the same group.

- 8 Explain what the atomic number tells us about an element.

- 9 Identify three elements that you think are named after famous people, and propose who that person was.

